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GASTLI: A Python package for coupled interior–atmosphere modelling of volatile-rich planets

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Summary

Gaseous exoplanets are far more diverse in mass, density and temperature than their Solar System counterparts. These range from temperate sub-Neptunes whose masses and sizes are below that of the Solar System ice giants, to extremely irradiated, Jupiter-mass exoplanets known as hot Jupiters. Their bulk composition is key to understanding their formation and evolution pathways, including their dominant formation mechanism (Thorngren et al., 2016) and interactions between the irradiation of the host star and the planetary atmosphere (Owen & Wu, 2013). Their building blocks consist of rocks and water (metals), and hydrogen and helium (H/He), which are present in a wide variety of mass fractions in the exoplanet population. For exoplanets that contain H/He, the planetary radius (or density) not only depends on mass, but also on the temperature due to the distance from its host star, its age, the mass of the core, and the fraction of H/He in its envelope. The inference of bulk composition from observational data – mass, radius, equilibrium temperature, age and atmospheric metallicity – requires non-trivial numerical methods that solve the equations of planetary interior structure. Additionally, interior models need to incorporate the effects of processes occurring in the planet atmosphere, such as radiative transfer, and the thermodynamical properties of the different materials, provided by the equation of state (EOS).

`GASTLI` (GAS gianT modeL for Interiors) is a Python package for interior structure modelling of volatile-rich exoplanets. The package combines a FORTRAN back end with a Python-based user interface. This enables fast convergence of the interior structure to a solution that fulfills the user input, while having a flexible and user-friendly interface. `GASTLI` was specifically designed to compute a wider range of combinations of core mass and H/He content in the envelope compared to existing open-source interior models. GASTLI also incorporates state-of-the-art thermodynamical data and EOS for rock, water and H/He. One of its key features is the coupling class, which couples the interior structure model to a grid of atmospheric temperature profiles to determine self-consistently the temperature of the outermost layer. GASTLI is released with a default atmospheric grid that assumes equilibrium chemistry and a clear atmosphere at varying internal temperatures computed with petitCODE (Mollière et al., 2015, 2017). Nonetheless, the flexible interface enables the user to use their own custom atmospheric grid, which is useful to explore the effect of the processes assumed in the upper atmosphere in the interior, such as the type of chemistry (equilibrium or disequilibrium), cloud implementations or 3D global circulation (GCM).

DRAFT

Statement of need

Interior structure codes are essential to carry out planet composition analyses and retrievals on mass and radius data. Two open-source interior structure models are currently available: MESA (Paxton et al., 2011, 2013, 2015, 2018, 2019) and MAGRATHEA (Huang et al., 2022). MESA is a well-known stellar interior structure software that has been applied to sub-Neptunes and gas giants (Chen & Rogers, 2016; Müller & Helled, 2021). The compositions of the envelope are limited to low metal contents due to the default EOSs, which is particularly limiting for planets with Neptune's mass or lower. Therefore, the implementation of alternative EOS in MESA are necessary, which is not straightforward for the user given the complexity of the code, and the input feature based on FORTRAN inlists. The computational time of one simple forward model is at least 1 minute. This renders MESA difficult to couple to atmospheric models, which can have a significant effect on the output of the interior calculations (Linder et al., 2019; Lopez & Fortney, 2014; Poser et al., 2019; Poser & Redmer, 2024). The second open-source planet interior model, MAGRATHEA, is a computationally fast code based on C++. Its layer structure (ideal gas EOS on top of liquid/ice water layer) makes its adaptation for applications to sub-Neptunes and gas giants a challenge to new users because these type of planets require the treatment of nonideal, high-pressure H/He envelopes.

The GASTLI Python package

The interior model module is contained in the `GASTLI.int_planet` class. The interior structure equations for hydrostatic equilibrium, Gauss's theorem for gravitational acceleration, convection, and mass conservation are solved by combining integration by the trapezoidal rule with an iteration scheme over all profiles (pressure, temperature, gravity and density). In each iteration, the trapezoidal integrations are carried for each point in a 1D grid that represents the radius, from the center of the planet to the outermost boundary of the top layer. The density is calculated by evaluating the EOS for rock (Lyon, 1992; Miguel et al., 2022), water (Mazevet et al., 2019) and H/He (Chabrier & Debras, 2021; Howard & Guillot, 2023) at the pressure and temperature conditions of each point in the spatial grid. The iteration sequence is considered to have converged when the difference between iterations is less than a given precision that can be modified by the user. The recommended and default value for this precision is 10^{-5}. The iteration scheme can also be stopped at a given number of iterations defined by the user, with a default of 30 maximum iterations, and a maximum possible value of 99 iterations. The boundary conditions to solve these equations include the top layer's pressure and temperature, which can be defined by the user. In addition to the profile convergence, the interior module ensures that the surface boundary conditions and the input mass are satisfied. A complete description of the interior module integration scheme can be found in Brugger (2018), and has been widely used (L. Acuña et al., 2021, 2022, 2023; Aguichine et al., 2021; Brugger et al., 2016, 2017; Mousis et al., 2020; Vivien et al., 2022). The new feature of the interior module includes the optimization of several of its routines, such as the trapezoidal integration and the evaluation of the EOS's density tables, and the parallelization of the calculation of the interior structure profiles with OpenMP in FORTRAN. One single computation of the interior module (function `calc_radius` in the `GASTLI.int_planet` class) for the default precision and number of iterations takes 1.89 seconds in an Apple M2 CPU, with parallelization on 8 cores (four at 3.49 GHz and four at 2.42 GHz processors).

The class `atm_models_interp` encloses the function that interpolates the atmospheric grid to obtain the temperature profile and interior boundary temperature. In addition, this class also provides the functions that calculate the atmospheric thickness and the atmosphere density profile. The latter is obtained by evaluating the EOS used by the interior module for H/He, and the AQUA water EOS (Haldemann et al., 2020). The atmospheric grid class and the interior module class are then combined in the `Coupling` class. This class self-consistently couples the interior module and the atmosphere grids to calculate the boundary temperature of

DRAFT

the outermost layer of the interior model. See L. Acuña et al. (2021) for a detailed description of the coupling algorithm, which has been used in previous work (L. Acuña et al., 2022, 2023; Aguichine et al., 2021). Finally, the class `Thermal_evolution` contains the functions necessary to evaluate a sequence of coupled interior-atmosphere models at different internal temperatures and solve the luminosity equation to obtain the internal temperatures and radii as a function of age (Fortney et al., 2013; Poser et al., 2019). The complete validation of GASTLI against mass-radius relationships obtained with interior models used in previous work (Fortney et al., 2013; Müller & Helled, 2021) can be found in Lorena Acuña et al. (2024).

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DRAFT